NORDDESIGN 2026 CONFERENCE
TAMPERE, FINLAND, 12-14 AUGUST 2026

Framework for Using Virtual Reality as a Tool in Service Design

Jonna Häkkilä[1], Emma Kirjavainen[1], Juri Etto[1], Kuisma Hurtig[1], Matilda Kalving[2], Siiri Paananen[1], Ashley Colley[1]

[1]University of Lapland
[2]Tampere University

Abstract: In this paper, we address the use of Virtual Reality (VR) as a tool in the service design process. We present a framework for using VR in the service design process, in which the VR-based and physical interaction phases are conducted sequentially to leverage the strengths of each method. VR is applied in contextualization, visualizing the potential, affirmation, and futurization phases, integrated into the arc of the service design process. The presented VRSD framework is evaluated in a service design case study that addresses the design of a recycling center, where VR was applied throughout the service design process. VR can be a powerful tool at all stages of service design, including ideating, iterating, and presenting, especially for saving time and enabling fast switching between designs. However, face-to-face meetings also remain important and play a key role in facilitating discussions and supporting the dynamics among designers.

Keywords: service design, Virtual Reality, design framework, design tools, co-design, 360 images, design process.

1 Introduction

In today's consumer world, products are increasingly often not just single items, but part of a larger product offering that relies on an ecosystem of services (Forlizzi, 2018). When conceptualizing and developing services, designers, developers, and service providers need to consider the holistic service experience and consumer interaction beginning prior to the actual product use and stretching all the way to post-use, including maintenance and the development of a long-term relationship with the customer. This has driven a new approach to design, Service Design. Service Design has rapidly emerged as a methodological approach that combines design thinking, user-centric design, and business orientation (Miettinen, 2016). Service design is now gaining increasing interest in the HCI research field, and especially user experience (UX) designers are at the forefront of this connection (Roto et al., 2021; Lee et al., 2022; Leinonen & Roto, 2023). Recent years have witnessed a growing number of case studies at the cross-section of HCI, UX, and service design (Chen et al., 2023; Häkkilä et al., 2023; Mazzarella et al., 2021; Wang & Dong, 2023).

When digital technologies are considered in conjunction with service design, they have mostly been an object of the design process. Indeed, HCI has predominantly used service design to refer to the design of digital services (Yap et al., 2021), or new technology has been seen as a driver of service innovation, with service design as a tool to leverage and improve innovation capabilities (Patrício et al., 2018). In our research, we consider technology as a tool and investigate its use during the service design process. Specifically, we address the use of immersive virtual reality (VR) to recreate servicescapes (Bitner, 1992) as part of the service design process.

In this paper, we introduce a design process framework for the use of virtual reality (VR) in service design. The framework was then evaluated in a service design process addressing the redesign of a recycling center service. Through the presented case study, we illustrate how we have utilized VR in service design, how the methodological framework can help structure its implementation in the design process, and discuss the strengths and limitations of the approach. Our research provides insight into how to successfully integrate VR as a new technological tool into the service design process and reports a novel methodological approach to service design with a holistic and systematic design process.

2 Related Work

To position our work, we first review prior work on service design, highlighting the key aspects of the approach that require support from design tools. Secondly, we discuss prior work that has used VR tools in the service design process.

2.1 Service Design

Service Design (SD) has been adopted as a new approach for consumer-centric service product creation during the past decade (Miettinen, 2016). Service design emphasizes a holistic approach, where the user's experience with the service is considered as a holistic arc or a journey rather than being considered piece by piece (Miettinen, 2016; Stickdorn et al., 2018). Service design ethos criticizes the conventional, unfortunately still common, way of designing services and product-

service systems, which has often occurred in silos, resulting in discontinuities in the user experience (Polaine et al., 2013, p. 22). In the service design approach, the user is seen to interact with the service through a set of service touchpoints, which are linked together in a service path. Service design highlights the consumer journey as a key concept (Følstad & Kvale, 2018; Yap et al., 2021), as it describes the consumer's actions, context, and the sequence of interactions with the service. Altogether, service design emphasizes the user's viewpoint and experience with the service and aims to look at the service through the user's eyes (Magnusson et al., 2003; Stickdorn et al., 2018).

Service design builds on the traditions of industrial design and user-centered design (Miettinen, 2016) and applies many methods that are derived from design and user research. Especially, user-centric and participatory design methods are applied. For instance, Curedale (2013) lists 250 methods for service designers to use in practice, many of which are familiar from the user-centered design tradition of HCI. Co-creation and participatory design methods are applied to involve different stakeholders, such as the service provider, consumers, and other relevant parties, to define the problem and suggest solutions as seeds for concept creation. Service prototyping methods are used to concretize the created development ideas into tangible, visual forms (Bae & Leem, 2014; Holmlid & Evenson, 2007). The creation of service prototypes is a central action in service design and focuses on stakeholders' co-creative experience, service process learnability, and iterative service development (Razek et al., 2019).

When connecting technology with service design, technology has mostly been considered as an element in the developed service, i.e., digital services. However, the relationship between the service design process itself and technology is not without challenges. Holmlid (2007) has claimed that when service design and interaction design meet, the service perspective poses a challenge for interaction design, and vice versa, technology use poses a challenge for service design. On the other hand, service design has been promoted as a useful approach when technical innovations are considered (Patrício et al., 2018). This dichotomy resembles the one faced by design research utilizing design practice, where the researcher is an active part in forming artifacts while keeping the research questions in mind (Fallman, 2008). Similarly, service designers can work with technology to harness its capabilities for further service innovations. Examples of this include, e.g., using large projected screens to simulate dynamic smart surfaces (Häkkilä et al., 2023) or using digital tools to keep diaries reflecting service journeys (Alhonsuo et al., 2022).

However, there is currently little research published on how service designers could actually use emerging technologies as tools during the service design process. Our research addresses this novel approach and contributes to the development of service design methodology.

2.2 VR as a Tool for Service Design

VR simulations potentially provide an effective design tool for evaluating products and services, as virtual artifacts can typically be created more easily than physical ones, and the virtual environment can be tightly controlled. There have been numerous prior works exploring the use of VR simulations as a service design tool (e.g., Bae & Leem, 2014; Boletsis, 2018; Boletsis et al., 2017; Colley et al., 2020), each typically based on a single service design case.

Boletsis et al. (2017) presented the use of VR tools in a service design process to enable 'virtual bodystorming'. The authors reported that the method was considered most suitable for prototyping services in open spaces (e.g., public transport, airports), but highlighted the importance of maintaining the service designer's role in facilitating the bodystorming process (Boletsis et al., 2017). Using the case example of a location-based guided audio tour, Boletsis (2018) developed a 'VR service walkthrough' method, reporting that it produced feedback of similar quality from test participants to a real-world-based method. Focusing on recreating a servicescape (Bitner, 1992) in VR, Kwon et al. (2015) used VR to evaluate customer experience in a simulated duty-free shop. In museum design, Pierroux et al. (2021) found value for researchers, museum experts, and architects in using collaborative VR tools for experimental museology, thereby combining interests across stakeholders. Whilst most prior work has focused solely on VR as a tool for presenting the service environment, Häfner et al. (2016) present a 'multi-dimensional' concept that visualizes additional layers of data, such as actors and artifacts.

Comparing the use of large-screen projection, VR, and AR as tools in service design, Colley et al. (2020) report that a head-mounted display worn by a co-design participant potentially reduces the interaction with the facilitating service designer, e.g., by preventing visibility of facial expressions. When exploring differences between real product experiences and simulated experiences in AR and VR, Voit et al. (2019) reported that the quantity and quality of the qualitative feedback elicited by a VR simulation were similar to those elicited by the original in situ physical experience. In a study exploring virtual fieldwork, Gushima and Nakajima note that VR methods were less effective in large outdoor locations than in restricted indoor spaces, as walking in a large space is difficult to experience in conventional VR (Gushima & Nakajima, 2021). This observation is supported by Bae and Leem (2014), who report positive results using a VR tool for prototyping interactions in a limited-space duty-free shop, but conflicts with Boletsis et al.'s (2017) recommendation of the approach for open spaces.

3 Framework for Using VR in the Service Design Process

Based on the literature review, a process framework has been synthesized (Figure 1). This distinguishes between the service design process phases with and without VR use. To synthesize the framework, we first identified the main functions for which VR was used, the stages of the process in which it was applied, and how the VR steps were integrated into the overall process.

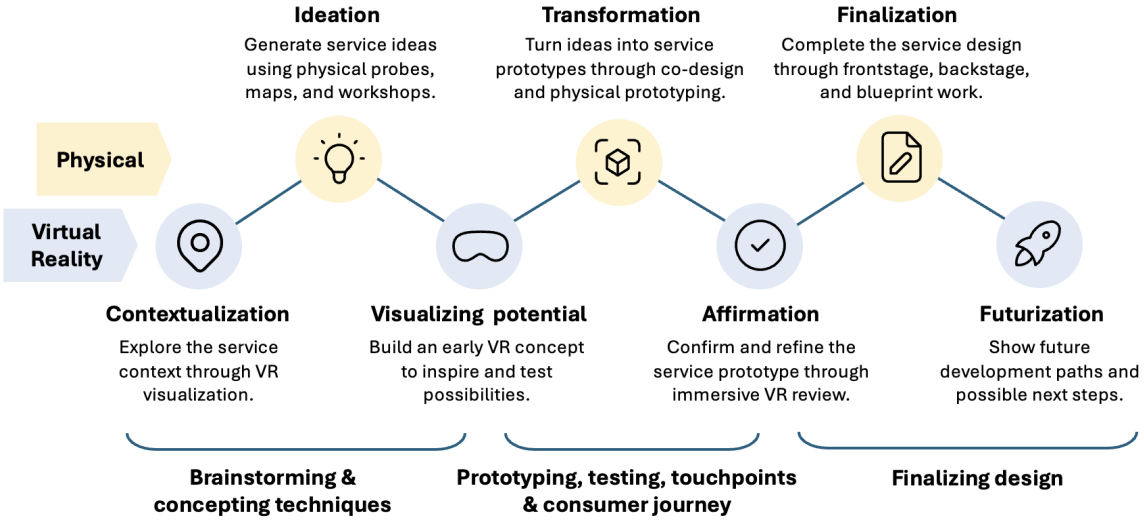


Figure 1. Framework for VR use in the service design process (VRSD) illustrating how VR-based and physical face-to-face phases alternate across the stages of the service design process.

3.1 Functions of VR

The benefits of using VR during service design include three main functions: contextualization, inspiration, and evaluation. In addition, a key benefit of VR is that it provides access to remote places, which the service design process may address, and enables people located remotely to participate in the design process, e.g., when multiple stakeholders in different locations are involved. Contextualizing the service the design process addresses is an important starting point for the design. Because of its immersive nature, VR can provide quite strong experiences and allow designers to almost see the context through the user's eyes. Due to its consistent presentation through a fixed simulation, VR can provide common ground for participants and stakeholders to discuss and begin ideating service development.

Much of the design work is about a creative process, and inspiration is an important element in setting the mindset, both in ideation and for finding alternative solutions when iterating the design. VR can be used as an inspirational tool by presenting imaginary futures and visualizing in an experience-rich way what the service could potentially be, see, e.g., (Simeone et al., 2022). Evaluating the service concept is important for identifying weak points and iterating on the design. A 3D-modeled service prototype offers an immersive way of presenting the design outcome. For instance, when physical spaces are designed as part of the service, VR can provide a full-body experience of how the space and its dimensions are perceived.

3.2 Framework for VR in Service Design

Prior art has shown that VR is not an optimal tool for all phases of the service design process. VR struggles, especially in the interactive and collaborative aspects of the design work. Whereas it can provide immersive visual experiences, it lacks tangibility, e.g., physical probe objects that co-design participants can handle. Also, the functions for recording comments on VR experiences are mostly still at the level of research prototypes, and iterating on the design of the virtual world during service design workshops and sessions is practically impossible. The inability to facilitate multi-user use cases for designers has been reported as a weakness of current VR tools, also in the context of design education (Häkkilä et al., 2018). These weaknesses are quite severe, as service design, by its nature, is highly collaborative and the process often relies on participatory design methods (Kuure et al., 2014). The benefits of physical interaction in these cases should not be neglected.

In our proposed framework for VR use in the service design process (VRSD), the phases of VR and physical interactions are conducted in turn to utilize the strengths of each method. In this framework, VR as a tool is applied in Contextualization, Visualizing the potential, Affirmation, and Futurization phases of the design process (Figure 1). Between the phases utilizing VR, there are design phases where physical interaction is conducted. This includes face-to-

face, or, as in pandemic times, online sessions (König et al., 2022; Tobin et al., 2020) with designers and other stakeholders. The phases of the design process, as presented in the framework (Figure 1), are the following:

- Contextualization (Virtual Reality, VR): Familiarizing with the service context through VR visualization
- Ideation (Physical): Utilizing brainstorming and concepting techniques conventionally used in service design, e.g., using physical probes, stakeholder maps, and creative workshops
- Visualizing the potential (VR): Creating an imaginary service concept in VR, utilizing ideation phase materials, to be used for inspiration and as the first version of the virtual service prototype
- Transformation (Physical): Transforming the initial concepts into a service prototype. Using conventional service design prototyping methods in order to take advantage of the qualities of co-design and physical prototyping. Identifying touchpoints and creating consumer journey descriptions.
- Affirmation (VR): Service prototype in VR as a tool for confirming the design. Design iterations are conducted using physical and face-to-face co-design workshop methods tailored to the needs, so that the final design can be presented in immersive VR.
- Finalization (Physical): Finalizing the design, with consideration of service front stage and backstage actions, creating the service blueprint
- Futurization (VR): Illustrating what could be the next development steps, providing seeds for future work after the project.

The proposed VRSD framework (Figure 1) is a methodological framework that offers researchers and design practitioners hands-on guidelines on when to use VR in the service design process. It can be used in conjunction with the existing design process frameworks that have gained popularity in service design. Here, especially the Double Diamond and Service Design Thinking models have established a position (Tschimmel, 2012). The Double Diamond design process framework, developed by the UK's Design Council in the early 2000s (Design Council, 2019), aims to provide a simplified, generic description of a design and innovation process, consisting of the phases Discover, Define, Develop, and Deliver. When mapped to the VRSD framework, it is seen that VR can be used in the Discover phase for contextualization, in the Define and Develop phases to visualize different design directions and their potential, to gain affirmation of the solutions, and in the Deliver phase to articulate the final service product. The Service Design Thinking model by Stickdorn and Schneider consists of four phases: Exploration, Creation, Reflection, and Implementation (Schneider & Stickdorn, 2011). Similarly, as with the Double Diamond design process model, the VRSD framework can be matched with the Exploration phase to contextualize the service design challenge, and in the Exploration and Creation phases by using VR to investigate and explore the potential of different solutions and to gain affirmation of the design decisions made. In the final Implementation phase, VR can help in articulating the final service product to customers and other stakeholders.

4 Validating the Framework - Service Design Case: Recycling Center

To validate the framework, the design process integrating VR was applied to a service design case, this time covering the whole design process (Figure 1). The selected case was the development of a new recycling area at a local recycling center. A set of service design workshops was arranged (Table 1), with a total of 21 different participants (13 self-identified as women and 8 as men) with a mean age of 32 years. Altogether, 8 participants attended the first workshop, 13 the second workshop, and 8 the third. Feedback was collected from each workshop and at the end of the whole design process.

Table 1. Workshops during the service design process in the validation case study

	Participants	Duration	How was VR used?
Workshop 1	5 company employees	2 hours	Viewing 360 images of the site to boost ideation
Workshop 2	13 potential users in groups of 3-4	45 minutes/group	Viewing and interacting with a 3D-modeled virtual recycling environment
Workshop 3	6 company employees	1 hour	Viewing and moving in the finalized, 3D-modeled concept of the recycling center

4.1 Steps of the Design Process

Before the first workshop, a meeting with the recycling company was held, and the planned recycling site area was captured using a 360-degree camera (Figure 2). At the beginning of the first service design workshop with company employees, participants were shown 360-degree images of the planned area via a head-mounted display (HMD). The area is quite remote, with no indoor spaces nearby, and the first workshop was held in the middle of a Nordic winter. Using VR allowed all participants to gain an equal understanding of the area and to ideate without venturing outdoors. Not all participants worked in the field; one participant said they had never visited the site in question (P2). The use of VR brought

the participants “on the same map”. Participants rated the effectiveness of VR in visualizing the area as 4.2 out of 5. Several participants mentioned how VR also made ideation easier by making the subject feel more concrete. VR provided a good foundation for the rest of the workshop, during which tools such as a visualized service path and area maps were projected onto a whiteboard. An initial concept for the recycling area was developed based on the results from the first workshop. The results included survey responses, visual materials from the workshop, and the facilitator’s observation notes. The recycling area concept was first 3D modeled in Blender and then made HMD-compatible using Unreal Engine. The virtual environment was interactive: participants could move around, pick up plastic waste items from the vehicle bed, and carry them to the appropriate recycling bins.



Figure 2. Actual site of the recycling area at the time of the first workshop (left), and the same place in the summertime with collected plastic rubbish (right).

The second workshop was held in late spring, and the virtual environment was set to summer weather conditions. Participants in this workshop were potential users of the recycling center who viewed the concept using a VR HMD (Figure 3). Participants found the ability to move around especially helpful for understanding the layout (P3). The interactive elements and realistic feel of the 3D space made the experience far more engaging than traditional methods such as image slideshows (P10, P11, P12). One participant (P11) also mentioned how the interactive elements in VR helped create a stronger memory imprint (compared to virtual environments, which you cannot interact with). Participants also commented positively on the ambient bird song incorporated into the VR experience.



Figure 3. VR simulation of the new recycling site and user testing.

Although modern VR HMD screens are high-quality, they still fall short of replicating the brightness, contrast, clarity, and other visual aspects of the real world as perceived by the human eye. Jin et al. (2021) have compared the differences between human eye imaging and HMD imaging and noted discrepancies. In this use case, this issue was observed during testing of signage in the virtual space. Several participants reported difficulty reading small text (P6, P11, P12). Even with these limitations, feedback was very positive, and participants rated VR’s help in visualizing the area at 4.76 out of 5. In the third workshop, the refined concept of the recycling area was presented to the company employees on their premises using a VR HMD (Figure 4). Participants took turns looking around at the finished virtual environment.

Overall, participants reported that the VR model was illustrative and provided useful ideas for developing the site. They also noted some things that would not work in practice. Overall, feedback for the use of VR was positive, with ratings of 4/5 for practicality and 4.5/5 for interest. However, the feedback was more critical than that given by the end user in the second workshop. The employees had better knowledge of the physical site, so their expectations for how closely the VR model mirrored reality were higher. For example, one participant (P3) pointed out that the ditches, roads, and lamp posts in the virtual model were not entirely faithful to reality. Such discrepancies may result in decreased perception of the value of the VR tool.



Figure 4. Employee viewing the final version of the recycling area through HMD (left). Designers involved in the case had an internal workshop reflecting the use of VR as a tool in service design (right).

4.2 Designers' Reflections on VR Use

To validate the framework from the designers' perspective and to collect feedback on VR use in the service design process, we organized a feedback workshop with five designers (3 identified as men and 2 as women) involved in the case study. One person familiar with the case but not involved in the design facilitated the workshop, which lasted for one hour. The framework was printed and displayed on the wall throughout the workshop, and the reflections were collected on post-it notes (Figure 4). During the workshop, the design process and the use of VR were discussed step by step in relation to the framework. The discussion was framed around four questions: 1) designers' views on the positive aspects of using VR in the service design process, 2) improvements that could be made, 3) the benefits of face-to-face work and physical presence during the design process, and 4) other comments on the framework.

The designers' reflections highlighted several benefits of using VR (Table 2). Mostly, the benefits of VR are related to two main themes: 1) saving time, and 2) the ability to modify and quickly change the design proposal. Saving time came up in different design phases. In Contextualization, using 360 photos and VR was seen as a quick way to present an overview of the design context, avoiding the need to travel to the physical location. In the articulation phase, 3D modeling for VR with the current tools was deemed fast, and changing between concept designs was easy, as there was "no need to run around changing signs in the field, just press a button" (Designer D#1). Organizing the evaluation workshop was expected to go more quickly when VR was used as a tool, as participants would otherwise have needed to commute to the field study location. Also, the use of VR was seen to ease recording participants' comments and to help organize participatory sessions, as organizing meetings and evaluation sessions in the wild is challenging, e.g., due to geographical distance and changing weather conditions. For instance, it was commented that "in the field, it is raining, heavy machines are running, and you try to collect feedback from the participant" (D#2). VR was also seen as beneficial when explaining the final design to the customer, as the presentation looked polished, was realistic, and the ability to move around in the virtual world enabled quick switching between locations and checking details. Using VR for futurization was noted to give the project a positive ending and an orientation towards the future.

Table 2. Designers' reflections on the benefits of VR in different phases of the design process according to the VRSD framework.

Design phase	Benefits of VR
Contextualization	Quick to familiarize with, saves time, no need for everyone to travel, provides common ground, inspiring for designers
Visualizing potential	Cost-efficient to prototype, easy to evaluate with users, easy to iterate, quick to switch between designs
Affirmation	Realistic way to present to customers, easy to give an overview of the area, inspiring for customers, fast to switch between locations, and view details
Futurization	Facilitates discussion on future development plans, inspiring customers

However, it was clearly seen that face-to-face interaction and in-person meetings around the same table were also necessary. There were two aspects to this. Firstly, face-to-face discussions were seen as important for collaboration and for facilitating the dynamics between the designers. Secondly, it was acknowledged that customers and user test participants primarily focused on the immersive experience while in VR. Although the VR simulation elicited immediate reactions, it was apparent that deeper discussions and reflections were more productive after the VR experience, when participants gathered around the same table and engaged in a collective conversation. Moreover, the lack of tools enabling collaboration around VR was seen as a limiting factor. In the discussions, VR was primarily seen as a feature for designers

working together, rather than a tool for involving external participants. Collaborative tools could enable designers to ideate and quickly test ideas together in the virtual world.

4.3 Summary of the Role of VR

Comparing these workshops highlights the diverse roles VR can play at different stages of the design process (Figure 5). In Workshop 1, including participants with differing levels of familiarity with the site, VR leveled the playing field for ideation and provided a shared reference point for early-stage ideation. Workshop 2 demonstrated VR's potential to make the design process more immersive for users who are less familiar with the site. Interaction in VR was also praised, making the experience more engaging. Finally, Workshop 3 showed that while VR is valuable for presenting and refining concepts, ensuring the accuracy of virtual representations is crucial, particularly for those with detailed knowledge of the physical site. In the end, the workshop with designers provided an overview of the benefits of using VR across each phase of the VRSD framework (Table 2).

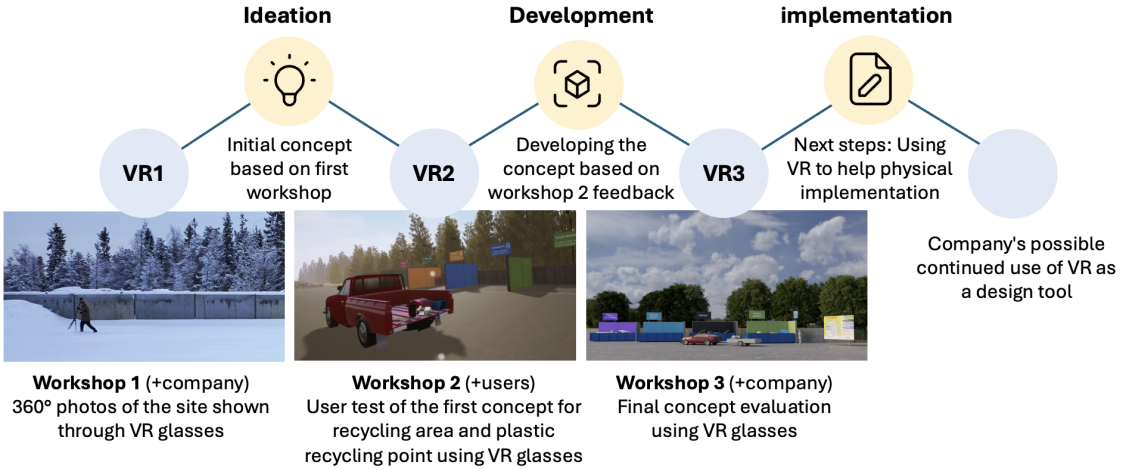


Figure 5. How the framework was applied in the presented service design case.

5 Discussion

This study demonstrates how VR can be integrated into the service design process, offering both opportunities and limitations. By employing VR across the different stages, from ideation to prototyping and validation, we explored its capacity to increase stakeholder engagement, simulate real-world environments, and create immersive service experiences. However, as with any emerging technology, VR's benefits must be weighed against its practical challenges, particularly in terms of collaboration and usability.

5.1 Opportunities and Limitations of VR in Service Design

Accessibility and immersion rise as clear benefits of using VR as a tool in service design. The service design process heavily utilizes stakeholder involvement in the design process. The ability to cross geographical distances with the help of immersive VR technologies is clearly an advantage. Especially for service design companies that operate over a larger area and with remote customers, the ability to smoothly engage distant stakeholders is a clear benefit. It has also been urged that design processes often need to consider a more complex contextual framework than traditionally considered in HCI design cases (Lee et al., 2022). VR can contribute to providing designers with a better tool for contextualizing the design with an immersive experience of the design context. This was also evident in the designers' comments in our validation case study. As a limitation of VR, the lack of collaborative aspects emerges as a clear disadvantage. This is not just about the technical difficulties of interacting and communicating in VR, where the features are still heavily underdeveloped. Once one puts on an HMD, virtual reality takes over, and communication and emotional connection between people co-located in the same physical space suffer. This is especially important to consider, as service design, as a field, relies heavily on participatory design and collaborative methods (Holmlid, 2012). The challenge of achieving a shared experience between co-located HMD users and non-users has been noted in HCI research (Gugenheimer et al., 2018). Moreover, difficulties in setting up VR collaborations have been recognized as a challenge for service design (Colley et al., 2020). This is especially notable if VR is implemented in the service design process to replace conventional methods or design setups. For example, setting up a monitor with a view of the VR experience for other participants can be a step into a more collaborative workflow. Sharing the VR experience with others allows them to see the viewpoint of the user, whether through a big shared screen or in a remote meeting. This can enable interesting design discussions and help different stakeholders understand each other's and users' perspectives. If the VR case demonstration is overly high-fidelity, it might take a lot of time and resources to develop and build. The trade-off between prototype fidelity, development agility, and cost is also noted by earlier research (Boletsis, 2018). Thus, simple 360 photos/videos or simple

content in a game engine should be considered, depending on the case. The time and effort spent on the VR application should be weighed and balanced. Whereas HMDs are already affordable technology, building VR environments takes certain skills and time. Even though accessing the VR simulation could be easily done by remote stakeholders, as already feasible in remote user studies and participant-owned HMDs (Rivu et al., 2021), the bottleneck to rapid, iterative design may lie in simulation development.

It has been suggested that conventional service prototypes are suitable for the early stage to roughly present a new service, whereas immersive service prototypes are beneficial for testing a new service scenario before implementation (Razek et al., 2019). We argue that the optimal process is more complex, with several phases where VR can be beneficial. On the other hand, there are stages in the design process where its limitations outweigh the potential benefits, and the conventional methods are more agile and efficient, and facilitate the group dynamics and discussions better.

5.2 Practical Implications of the VRSD Framework

The presented VRSD framework aims to provide a practical, methodological tool that defines the different phases of the service design process and enables VR integration as a useful tool. As the service design process begins with familiarizing oneself with the context, studying the user viewpoint, and exploring the topic, VR can add value by contextualizing the design task at the beginning of the process. In the next step of the design process, i.e., in the ideation phase, brainstorming and concepting techniques are used. However, in this phase, it is more beneficial to focus on physical methods and favor a face-to-face setting, where the much-used collaborative, co-design, and rapid prototyping methods are more easily applicable than in VR. When prototyping iterations and touchpoints, VR can help visualize the potential of different solutions and, at the same time, affirm the selected choices and design decisions. When the final outcome of the service is presented, VR can help the service design team articulate the service product to customers and other stakeholders in an immersive way. With these steps, VR can also be integrated into existing, commonly used design process frameworks, such as the Double Diamond (Design Council, 2019) and Service Design Thinking (Schneider & Stickdorn, 2011) models. This also bridges the gap between HCI and service design, as HCI studies, such as the use of virtual environments in user studies (Mäkelä et al., 2020; Rivu et al., 2021), can be integrated into the service design process, for instance, in the iterative design and affirmation phases. As a practical implementation, the VRSD framework provides a step-by-step guide for integrating VR into the service design process. Whereas in the case presented in this paper, VR was used primarily as a tool to address practical needs that arose, not as predefined or planned steps in the overall service design process, the VRSD framework offers a model for more robust planning and formulation of the service design process. This can support practitioners, for instance, in planning resources and scheduling different design phases. Recognizing that VR is a useful tool in some phases but that physical methods and face-to-face interaction are preferred in others helps plan workshops and meetings where designers and other stakeholders gather, sometimes over long geographical distances.

5.3 Limitations

We note that our validation study was designed to integrate the steps into the VRSD framework for the service design process and that systematic feedback was collected throughout the case study. Moreover, we wish to emphasize that our research focuses on service design, not controlled user research or evaluation with user tests, and covers the holistic process of a real-world service design case.

6 Conclusion

In this paper, we have addressed the use of VR as a tool for service design by developing a framework that was validated through its application to a service design case. VR can be used to provide contextualization, inspiration, and evaluation, all of which are important aspects of the service design process. We conclude that the strengths of VR are especially in providing access to remote places, which the service design process addresses, and in enabling people in remote locations to participate in the design process when stakeholders are distributed across different locations. Using VR tools supports the iterative nature of many design processes and can be applied at various stages. The use of VR, however, is not without limitations. With VR, there are severe challenges in applying co-creation aspects, and many common methods used in conventional service design processes, such as role-playing, tangible low-fidelity prototyping, stakeholder mapping with post-it notes, and other ideation and co-design techniques, which are straightforward when participants are sitting around the same table, are lacking. We have presented and validated a methodological framework that illustrates in which phases of the service design process VR is beneficial and where, on the other hand, physical-world methods are more suitable. In the VRSD framework, we frame the use of VR through four phases: Contextualization of the research problem, Visualizing the potential of different concept solutions, Affirmation to test and validate the service concept, resulting in the final design, and finally, Futurization is conducted at the end to provide seeds for potential future work. Our work contributes to the development of service design methodology by utilizing emerging technologies. In addition to the research contribution, the framework provides hands-on guidance for service design and HCI practitioners by providing a framework for the design process planning. It helps balance between VR and physical methods across different phases of the design process, thereby optimizing the service design process steps and the outcome.

Acknowledgments

This study was funded by the Rethinking Plastics in a Sustainable Circular Economy (PlastLIFE) project (LIFE21-IPE-FI-PlastLIFE). The PlastLIFE project was co-funded by the European Union. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the European Union or CINEA. Neither the European Union nor the granting authority can be held responsible for them.

References

Alhonsuo, M., Sarantou, M., Miettinen, S., 2022. Digital Storytelling to Share Service Experiences and Find Insights into Health Care, in: Pfannstiel, M.A., Brehmer, N., Rasche, C. (Eds.), Service Design Practices for Healthcare Innovation: Paradigms, Principles, Prospects. Springer, Cham, pp. 529–547. https://doi.org/10.1007/978-3-030-87273-1_26

Bae, D.J., Leem, C.S., 2014. A Visual Interactive Method for Service Prototyping. Managing Service Quality: An International Journal 24(4), 339–362. https://doi.org/10.1108/MSQ-12-2013-0281

Bitner, M.J., 1992. Servicescapes: The Impact of Physical Surroundings on Customers and Employees. Journal of Marketing 56(2), 57–71. https://doi.org/10.1177/002224299205600205

Boletsis, C., 2018. Virtual Reality for Prototyping Service Journeys. Multimodal Technologies and Interaction 2(2), 14. https://doi.org/10.3390/mti2020014

Boletsis, C., Karahasanovic, A., Fjuk, A., 2017. Virtual Bodystorming: Utilizing Virtual Reality for Prototyping in Service Design, in: De Paolis, L.T., Bourdot, P., Mongelli, A. (Eds.), Augmented Reality, Virtual Reality, and Computer Graphics. Lecture Notes in Computer Science, vol. 10324. Springer, Cham, pp. 279–288. https://doi.org/10.1007/978-3-319-60922-5_22

Chen, L., Huang, M., Liu, Z., Jiang, Y., Wu, T., 2023. Applying Service Design Thinking to UX Research: A Case of Smart Campus Dance Experience Design, in: Marcus, A., Rosenzweig, E., Soares, M.M. (Eds.), Design, User Experience, and Usability. Lecture Notes in Computer Science, vol. 14031. Springer, Cham, pp. 57–74. https://doi.org/10.1007/978-3-031-35696-4_5

Curedale, R., 2013. Service Design: 250 Essential Methods. Design Community College Inc., Topanga, CA.

Colley, A., Suoheimo, M., Häkkilä, J., 2020. Exploring VR and AR Tools for Service Design, in: Proceedings of the 19th International Conference on Mobile and Ubiquitous Multimedia. ACM, pp. 309–311. https://doi.org/10.1145/3428361.3432074

Design Council, 2019. The Double Diamond: A Universally Accepted Depiction of the Design Process. Design Council.

Fallman, D., 2008. The Interaction Design Research Triangle of Design Practice, Design Studies, and Design Exploration. Design Issues 24(3), 4–18. https://doi.org/10.1162/desi.2008.24.3.4

Følstad, A., Kvale, K., 2018. Customer Journeys: A Systematic Literature Review. Journal of Service Theory and Practice 28(2), 196–227. https://doi.org/10.1108/JSTP-11-2014-0261

Forlizzi, J., 2018. Moving Beyond User-Centered Design. interactions 25(5), 22–23. https://doi.org/10.1145/3239558

Gugenheimer, J., Stemasov, E., Frommel, J., Rukzio, E., 2018. A Demonstration of ShareVR: Co-Located Experiences for Virtual Reality Between HMD and Non-HMD Users, in: 2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE, pp. 755–756. https://doi.org/10.1109/VR.2018.8446551

Gushima, K., Nakajima, T., 2021. Virtual Fieldwork: Designing Augmented Reality Applications Using Virtual Reality Worlds, in: Chen, J.Y.C., Fragomeni, G. (Eds.), Virtual, Augmented and Mixed Reality. Lecture Notes in Computer Science, vol. 12770. Springer, Cham, pp. 417–430. https://doi.org/10.1007/978-3-030-77599-5_29

Jin, Y., Zhou, X., Xiao, W., Li, J., Xue, C., 2021. Comparison of Differences Between Human Eye Imaging and HMD Imaging, in: 2021 5th International Conference on Artificial Intelligence and Virtual Reality (AIVR). ACM, pp. 72–76. https://doi.org/10.1145/3480433.3480434

Häfner, P., Elstermann, M., Häfner, V., Ovtcharova, J., 2016. Virtual Reality Based Rapid Service Prototyping Tool, in: Proceedings of EuroVR Conference 2016, Athens, Greece.

Häkkilä, J., Colley, A., Väyrynen, J., Yliharju, A.-J., 2018. Introducing Virtual Reality Technologies to Design Education. Seminar.net 14(1), 1–12. https://doi.org/10.7577/seminar.2584

Häkkilä, J., Löchtefeld, M., Brun, D., Colley, A., 2023. Service Design Methods for the Design of Smart Surfaces, in: Hamid, U.Z.A., Suoheimo, M. (Eds.), Service Design for Emerging Technologies Product Development: Bridging the Interdisciplinary Knowledge Gap. Springer Series in Design and Innovation, vol. 29. Springer, Cham, pp. 235–247. https://doi.org/10.1007/978-3-031-29306-1_13

Holmlid, S., 2007. Interaction Design and Service Design: Expanding a Comparison of Design Disciplines, in: Nordes 2007: Design Inquiries, 2nd Nordic Design Research Conference, Stockholm. https://doi.org/10.21606/nordes.2007.031

Holmlid, S., 2012. Participative; Co-Operative; Emancipatory: From Participatory Design to Service Design, in: Conference Proceedings ServDes.2009; DeThinking Service; ReThinking Design. Linköping Electronic Conference Proceedings 59, 105–118.

Holmlid, S., Evenson, S., 2007. Prototyping and Enacting Services: Lessons Learned from Human-Centered Methods, in: Proceedings from the 10th Quality in Services Conference, QUIS 10, Orlando, Florida.

König, A., Hatzakis, T., Andrushevich, A., Hoogerwerf, E.-J., Vasconcelos, E., Launo, C., Alčiauskaitė, L., Barbosa, S., Andersen, K., 2022. A Reflection on Participatory Research Methodologies in the Light of the COVID-19 – Lessons Learnt from the European Research Project TRIPS. Open Research Europe 1, 153. https://doi.org/10.12688/openreseurope.14315.2

Kuure, E., Miettinen, S., Alhonsuo, M., 2014. Change Through Service Design – Service Prototyping as a Tool for Learning and Transformation, in: Lim, Y., Niedderer, K., Redström, J., Stolterman, E., Valtonen, A. (Eds.), Design's Big Debates: DRS International Conference 2014. Umeå Institute of Design, Umeå University, Umeå, pp. 469–482.

Kwon, R.-H., Kim, K.-J., Kim, K.-H., Hong, Y.-S., Kim, B., 2015. Evaluating Servicescape Designs Using a VR-Based Laboratory Experiment: A Case of a Duty-Free Shop. Journal of Retailing and Consumer Services 26, 32–40. https://doi.org/10.1016/j.jretconser.2015.05.001

Lee, J.-J., Yap, C.E.L., Roto, V., 2022. How HCI Adopts Service Design: Unpacking Current Perceptions and Scopes of Service Design in HCI and Identifying Future Opportunities, in: CHI Conference on Human Factors in Computing Systems. ACM, Article 530, pp. 1–14. https://doi.org/10.1145/3491102.3502128

Leinonen, A., Roto, V., 2023. Service Design Handover to User Experience Design – A Systematic Literature Review. Information and Software Technology 154, 107087. https://doi.org/10.1016/j.infsof.2022.107087

Magnusson, P.R., Matthing, J., Kristensson, P., 2003. Managing User Involvement in Service Innovation: Experiments with Innovating End Users. Journal of Service Research 6(2), 111–124. https://doi.org/10.1177/1094670503257028

Mäkelä, V., Radiah, R., Alsherif, S., Khamis, M., Xiao, C., Borchert, L., Schmidt, A., Alt, F., 2020. Virtual Field Studies: Conducting Studies on Public Displays in Virtual Reality, in: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. ACM, pp. 1–15. https://doi.org/10.1145/3313831.3376796

Mazzarella, F., May, A., Mitchell, V., 2021. A Methodological Framework for Crafting Situated Services. Journal of Service Management 32(5), 752–782. https://doi.org/10.1108/JOSM-05-2020-0188

Miettinen, S., 2016. Introduction to Industrial Service Design, in: Miettinen, S. (Ed.), An Introduction to Industrial Service Design. Routledge, Abingdon, pp. 3–14. https://doi.org/10.4324/9781315566863

Patrício, L., Gustafsson, A., Fisk, R., 2018. Upframing Service Design and Innovation for Research Impact. Journal of Service Research 21(1), 3–16. https://doi.org/10.1177/1094670517746780

Pierroux, P., Sauge, B., Steier, R., 2021. Exhibitions as a Collaborative Research Space for University-Museum Partnerships, in: Achiam, M., Haldrup, M., Drotner, K. (Eds.), Experimental Museology: Institutions, Representations, Users. Routledge, pp. 149–166. https://doi.org/10.4324/9780367808433

Polaine, A., Løvlie, L., Reason, B., 2013. Service Design: From Insight to Implementation. Rosenfeld Media, Brooklyn, NY.

Razek, A.R.A., van Husen, C., Pallot, M., Richir, S., 2019. Comparing Conventional Versus Immersive Service Prototypes: An Empirical Study. International Journal of Virtual Reality 19(2), 16–34. https://doi.org/10.20870/IJVR.2019.19.2.2914

Rivu, R., Mäkelä, V., Prange, S., Rodriguez, S.D., Piening, R., Zhou, Y., Köhle, K., Pfeuffer, K., Abdelrahman, Y., Hoppe, M., et al., 2021. Remote VR Studies: A Framework for Running Virtual Reality Studies Remotely via Participant-Owned HMDs. ACM Transactions on Computer-Human Interaction 28(6), Article 46, pp. 1–36. https://doi.org/10.1145/3472617

Roto, V., Mitchell, V., Cockbill, S., Lee, J.-J., Lai-Chong Law, E., Zimmerman, J., 2021. Introduction to Service Design for UX Designers, in: Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems. ACM, Article 148, pp. 1–3. https://doi.org/10.1145/3411763.3444999

Schneider, J., Stickdorn, M., 2011. This Is Service Design Thinking: Basics, Tools, Cases. Wiley.

Simeone, A.L., Cools, R., Depuydt, S., Gomes, J.M., Goris, P., Grocott, J., Esteves, A., Gerling, K., 2022. Immersive Speculative Enactments: Bringing Future Scenarios and Technology to Life Using Virtual Reality, in: CHI Conference on Human Factors in Computing Systems. ACM, Article 17, pp. 1–20. https://doi.org/10.1145/3491102.3517492

Stickdorn, M., Hormess, M.E., Lawrence, A., Schneider, J., 2018. This Is Service Design Doing: Applying Service Design Thinking in the Real World. O'Reilly Media, Sebastopol, CA.

Tobin, C., Mavrommati, G., Urban-Rich, J., 2020. Responding to Social Distancing in Conducting Stakeholder Workshops in COVID-19 Era. Societies 10(4), 98. https://doi.org/10.3390/soc10040098

Tschimmel, K., 2012. Design Thinking as an Effective Toolkit for Innovation, in: Proceedings of the XXIII ISPIM Conference: Action for Innovation: Innovating from Experience. ISPIM, Barcelona, pp. 1–20. https://doi.org/10.13140/2.1.2570.3361

Voit, A., Mayer, S., Schwind, V., Henze, N., 2019. Online, VR, AR, Lab, and In-Situ: Comparison of Research Methods to Evaluate Smart Artifacts, in: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. ACM, Article 406, pp. 1–14. https://doi.org/10.1145/3290605.3300737

Wang, X., Dong, J., 2023. Research on the Strategy of Travel Service System for Visually Impaired People Based on Service Design Thinking, in: Proceedings of the Eleventh International Symposium of Chinese CHI. ACM, pp. 25–36. https://doi.org/10.1145/3629606.3629610

Yap, C.E.L., Lee, J.-J., Roto, V., 2021. How HCI Interprets Service Design: A Systematic Literature Review, in: Ardito, C., Lanzilotti, R., Malizia, A., Petrie, H., Piccinno, A., Desolda, G., Inkpen, K. (Eds.), Human-Computer Interaction – INTERACT 2021. Lecture Notes in Computer Science, vol. 12933. Springer, Cham, pp. 259–280. https://doi.org/10.1007/978-3-030-85616-8_16